

Student Guide — middle / module-07-eleven-multiplication

IKS Curriculum

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Student Workbook

Student Workbook — Module 7 Eleven Multiplication (Middle)

Name: _____ **Class:** _____ **Section:** _____

This is your workbook for the next two weeks. Bring it to every class.

Day 1 — Baseline

My baseline time for 73×11 (long multiplication): _____ seconds

My first try with the $\times 11$ trick: _____ seconds (mostly worked? Y / N)

The trick (in my own words): 1. _____ 2. _____
_____ 3. _____

Day 2 — The Carry Case

When the digit-sum is ≥ 10 : - Middle digit = _____ - Carry the _____ into the _____

Worked example (mine): _____ $\times 11$ = _____ (digit-sum = _____)

Day 3 — Why It Works

The algebra (in my handwriting):

$$\begin{aligned}
 (10a + b) \times 11 &= && \text{(distribute)} \\
 &= && \text{(rearrange)} \\
 &= 100a + 10(a+b) + b && \text{(final form)}
 \end{aligned}$$

In one sentence: the trick works because _____

Day 4 — Three-Digit $\times 11$

The pattern for $abc \times 11$: _____

Worked example: 234×11

Thousands: ____ Hundreds: ____ Tens: ____ Units: ____ $\rightarrow$ _____

Day 5 — Mental-Math Speed Day

Day 1 baseline (73×11 , long way): ____ sec Day 5 retest (73×11 , with trick): ____ sec %
improvement: _____

My project choice: _____ (video / workbook / proof poster) Partner: _____

Day 6 — *Anurūpyeṇa*

For 2-digit $ab \times (10 + k)$: - units = _____ - tens = _____
 _____ - hundreds = _____

Example: $45 \times 13 =$ _____

Day 7 — When the Trick Doesn't Help

The $\times 11$ trick (and its scaled cousins) works for multipliers of the form _____.

For multipliers far from 10, use _____.

Day 8 — Tirthaji's Sūtra System

Vedic Mathematics (Tirthaji) was published in the year ____ (____ years after Tirthaji died).

Tirthaji claimed the sūtras come from an _____ (a Vedic appendix text).

The mathematician _____ (Dani / Plofker / ?) wrote a critique in _____.

My current view: _____

Day 9 — Project Progress

What I worked on today: _____

What's still left: _____

Day 10 — Reflection

1. What was the most surprising thing you learned about $\times 11$?

2. Did your view of “Vedic mathematics” change during this module? How?

3. If you had three more days, what would you do differently?

Glossary (your own)

Sanskrit	Meaning	First met
<i>Ekādhikena Pūrveṇa</i>	“by one more than the previous”	Day 1
<i>Anurūpyeṇa</i>	“proportionally”	Day 6
<i>Tirthaji</i>	name (1884–1960)	Day 8
<i>sūtra</i>	aphorism / formula	Day 8
<i>Parīśiṣṭa</i>	appendix / supplement	Day 8

My favourite $\times 11$ problem from the module

(Write the problem and your worked answer. Use the dot trick to show off.)

Problem: _____ $\times 11 =$ _____

Why I like it: _____

Day 1 — The Hook: $2_3 = 253$

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute any 2-digit $\times 11$ product where the digit-sum is less than 10, in under 3 seconds, and will have recorded a personal baseline time for the same problem using standard long multiplication.

Materials

- Printed baseline sheet: ONE multiplication problem (73×11). One per student.
- Stopwatch or wall clock with a second hand. (One per pair is plenty.)
- Whiteboard / large paper to demonstrate.

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: ekādhikena pūrveṇa; lit. “by one more than the previous”) — the Sanskrit name Tirthaji (1965) gives the underlying sūtra. We use it as the cultural anchor; the technique is what students take home.

Lesson flow

Warm-up (5 min)

- Teacher: “How long does it take you to compute 73×11 the normal way? Make a guess.”
- Take 3–4 guesses. Write the range on the board.

Baseline timing (8 min)

- Hand out the baseline sheet face-down. Students write their name and predicted time on the back.
- On “Go,” flip and solve 73×11 using long multiplication.
- Record actual time honestly. Mark answer themselves.
- Expected range: 15–60 seconds.

Teacher demonstration (12 min) — The trick

- Write on the board:

$$73 \times 11 = ?$$

- “I’m going to show you a method that takes about 2 seconds, then we’ll figure out why it works.”

Step 1: Write the digits, with a gap: 7 _ 3

Step 2: Add the two digits: $7 + 3 = 10$.

Wait – that’s the carry case. Park it.

- *Reset. Different number first. Try 23×11 :*

Step 1: Split: 2 _ 3

Step 2: Add: 2 + 3 = 5

Step 3: Drop in: 2 5 3 → 253.

Check on the board with long multiplication: $23 \times 11 = 23 \times 10 + 23 = 230 + 23 = 253$. ✓

- Try 45×11 :

4 _ 5 → 4 + 5 = 9 → 4 9 5 → 495.

- Try 62×11 :

6 _ 2 → 6 + 2 = 8 → 6 8 2 → 682.

- “Pattern: outer digits stay the same; middle digit = their sum.”
- Now name it. “Tirthaji (1965) calls this ***Ekādhikena Pūrveṇa*** — ‘by one more than the previous.’ That is the Sanskrit name of the *sūtra*. The trick itself is just $(10a + b) \times 11 = 100a + 10(a+b) + b$, which we’ll prove on Day 3.”

Activity (15 min) — Try it yourself

- Worksheet (handed out): 10 two-digit $\times 11$ problems, ALL with digit-sum < 10 .

12×11	23×11	34×11	45×11	52×11
61×11	16×11	27×11	35×11	44×11

- Pair work. Partner times each other. Goal: each problem under 3 seconds.
- After 10 min, students re-do 73×11 on the back of the baseline sheet. *Hmm — $7 + 3 = 10$. That’s tomorrow’s lesson.*

Wrap-up (5 min)

- Show of hands: who finished all 10 in under 30 seconds? (Most will.)
- Tease tomorrow: “ 73×11 broke the rule because $7 + 3 = 10$ and you can’t write 10 in the middle of a number. Tomorrow — the carry case.”

Student-facing content

The $\times 11$ Trick (no-carry case)

For any 2-digit number with digit-sum < 10 : 1. **Split** the digits with a gap. 2. **Add** the two digits. 3. **Drop** the sum into the gap.

Example: $34 \times 11 \rightarrow 3_4 \rightarrow 3+4 = 7 \rightarrow 374$.

Tirthaji (1965) names the underlying sūtra ***Ekādhikena Pūrveṇa*** — “by one more than the previous.” The trick is older than its modern Sanskrit name; the algebra (which we’ll do on Day 3) is the same in every language.

Homework

- Find any two 2-digit numbers (digit-sum < 10) in your house — phone numbers, page numbers, prices. Compute each $\times 11$ using the trick. Bring 5 worked examples to class tomorrow.

Differentiation

- Tier up:** Try 3-digit $\times 11$ (e.g. 234×11 , 152×11). See if you can guess the pattern before Day 4.
- Tier down:** If the gap is confusing, write the answer as a 3-digit number directly: hundreds = first digit, tens = sum, units = second digit.

Teacher notes

- The baseline time matters.** Day 5’s retest depends on having a Day 1 record.
- Don’t dwell on the Sanskrit name today. *Ekādhikena Pūrveṇa* is the cultural anchor; the win is mental speed.
- Some students will already know the trick (especially those from competitive-math backgrounds). Have them be peer-coaches. Reframe their advantage as “you’ll learn the WHY this week.”
- About 30% of randomly-chosen 2-digit numbers have digit-sum ≥ 10 (the carry case). So today’s worksheet hand-picks the no-carry ones; tomorrow we add the carries.

Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. Sūtra 1, *Ekādhikena Pūrveṇa*.

Day 2 — The Carry Case

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will correctly handle the carry case in 2-digit $\times 11$ (where the digit sum ≥ 10), including problems like $87 \times 11 = 957$, in under 5 seconds.

Materials

- Whiteboard.

- Printed worksheet of 12 problems, mixing no-carry and carry cases.
- Stopwatch.

Lesson flow

Warm-up (5 min)

- Neighbour-sum drill: teacher calls a 2-digit number with digit-sum < 10 ; class chorus the $\times 11$ product.
- “12 times 11!” $\rightarrow$ “132!” — “34 times 11!” $\rightarrow$ “374!” — 10 in a minute.

Recall (3 min)

- Yesterday’s problem: 73×11 . $7 + 3 = 10$. Why was this awkward?
- “You can’t put 10 in the middle of a 3-digit number. So what do we do? Carry the 1 — exactly like in standard addition.”

Core (15 min) — The carry case

1. Demonstrate 73×11 fully:

Step 1: Split: $7 \quad _ \quad 3$

Step 2: Add: $7 + _ 3 = 10$

Step 3: Write the units digit of the sum (0) in the middle: $7 \quad 0 \quad 3$

Step 4: Carry the tens digit of the sum (1) into the leftmost digit: $7 + 1 = 8$.

Step 5: Answer: $8 \quad 0 \quad 3 \rightarrow 803$.

Check: $73 \times 11 = 73 \times 10 + 73 = 730 + 73 = \mathbf{803}$. ✓

2. Try 87×11 :

$8 \quad _ \quad 7$

$8 + _ 7 = 15$

Middle = 5; carry 1 into the 8 $\rightarrow$ 9.

Answer: $9 \quad 5 \quad 7 \rightarrow 957$.

Check: $87 \times 11 = 870 + 87 = \mathbf{957}$. ✓

3. Try 99×11 :

$9 \quad _ \quad 9$

$9 + _ 9 = 18$

Middle = 8; carry 1 into the 9 $\rightarrow$ 10.

But 10 isn't a single digit — so it becomes a "1" carried further: leftmost = 0, with a 1 carried to the left of that.

Answer: $1 \quad 0 \quad 8 \quad 9 \rightarrow 1089$.

Check: $99 \times 11 = 990 + 99 = \mathbf{1089}$. ✓ (4-digit answer because the carry cascaded.)

4. The full procedure (boxed on the board):

For any 2-digit number $ab \times 11$:

1. Compute $a + b$.
2. If the sum is < 10 : answer is $a (a+b) b$.
3. If the sum is ≥ 10 : middle digit = units digit of sum; add 1 to a .
4. If the new a is ≥ 10 (only happens at 99): cascade the carry.

Activity (17 min) — Mixed practice

- Worksheet of 12 problems, half no-carry, half carry:

21×11	35×11	47×11	58×11
62×11	74×11	83×11	89×11
91×11	99×11	18×11	27×11

- Round 1 (8 min): solo, untimed. Self-check with long multiplication on any 3 of them.
- Round 2 (8 min): paired. Partner calls a problem; you answer aloud. Goal: under 5 seconds per problem.

Wrap-up (5 min)

- Quick chorus: “*What does the trick break into?*” — Split, add, drop (and carry if needed).
- Tease Day 3: “*Tomorrow we PROVE why this works. You’ll write the algebra yourself.*”

Student-facing content

The $\times 11$ Trick (with carry)

1. Split the digits.
2. Add them.
3. **If the sum < 10 :** drop it in the middle. Done.
4. **If the sum ≥ 10 :** keep the units digit of the sum as the middle; add 1 to the leftmost digit.

Examples: - 45×11 : $4+5=9 \rightarrow 495 = \mathbf{495}$ - 87×11 : $8+7=15 \rightarrow 8(\mathbf{1})57 \rightarrow \mathbf{957} = \mathbf{957}$ -
 99×11 : $9+9=18 \rightarrow 9(\mathbf{1})89 \rightarrow \text{cascade} \rightarrow \mathbf{1089} = \mathbf{1089}$

Homework

- 8 mental $\times 11$ problems (carry case): 76, 84, 85, 88, 93, 96, 67, 78 — multiplied by 11. Write the answers.
- Time yourself. Goal: under 30 seconds total.

Differentiation

- **Tier up:** Find a 2-digit number where the trick gives a 4-digit answer (i.e. $99 \times 11 = 1089$). Can you find any others, or is 99 unique? Justify.
- **Tier down:** Solve only the no-carry problems on the worksheet today. We’ll cement carries on Day 4.

Teacher notes

- The most common error: students remember to write the units digit of the middle sum but **forget** to add the carry into the leftmost digit. Watch for “ $73 \times 11 = 703$ ” with no carry added — this is a classic mistake.
- The 99×11 case (and the cascading carry) genuinely surprises students. Lean into it — it’s evidence the algebra is doing real work, not arithmetic by coincidence.
- If a student asks “why does this work?” — that’s Day 3. Park it. Honour the curiosity by writing the question on a board sticky.

Citation(s) used in this lesson

- Tirthaji (1965), sūtra 1 (*Ekādhikena Pūrveṇa*) — the technique name.
- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center) — worked examples of the carry case.

Day 3 — Why It Works

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will derive the identity $(10a + b) \times 11 = 100a + 10(a + b) + b$ and explain in their own words why the $\times 11$ trick is not a coincidence.

Materials

- Whiteboard.
- Worksheet: “explain to a friend” prompt with space for student algebra.

Lesson flow

Warm-up (5 min)

- Neighbour-sum chorus: 15 quick $\times 11$ problems, mix of carry and no-carry.
- “*By now your hands know the trick. Today your brain catches up.*”

Core (20 min) — The algebra

1. **Write any 2-digit number as $10a + b$.** Make sure every student gets this.

$$\begin{array}{lcl} 23 = 10 \cdot 2 + 3 & \rightarrow & a = 2, b = 3 \\ 87 = 10 \cdot 8 + 7 & \rightarrow & a = 8, b = 7 \end{array}$$

2. **Distribute the multiplication by 11.** Step by step on the board, students copy:

$$\begin{aligned} (10a + b) \times 11 & \\ &= (10a + b) \times (10 + 1) \\ &= (10a + b) \times 10 + (10a + b) \times 1 \\ &= 100a + 10b + 10a + b \end{aligned}$$

$$= 100a + 10a + 10b + b$$

$$= 100a + 10(a + b) + b$$

3. **Read the final line out loud.** “ $100a + 10(a+b) + b$. What does this say in plain language?”

- $100a$ — the **hundreds** digit is a (the original first digit).
- $10(a+b)$ — the **tens** digit is $a + b$ (the digit sum).
-

- b — the **units** digit is b (the original second digit).

“That IS the trick. The algebra spells out: first digit, sum, last digit.”

4. **Check on 23:**

$$a = 2, b = 3.$$

$$100 \cdot 2 + 10 \cdot 5 + 3 = 200 + 50 + 3 = 253. \checkmark$$

5. **Now the carry case** — where does the carry come from?

$$a = 8, b = 7.$$

$$100 \cdot 8 + 10 \cdot 15 + 7 = 800 + 150 + 7 = 957.$$

“Notice: $10 \times 15 = 150 = 100 + 50$. The 100 contribution slides into the hundreds column — that IS the carry. So the algebra ALSO explains why we add 1 to the leftmost digit when the sum is ≥ 10 .”

Activity (15 min) — Explain to a friend

- Worksheet prompt:

Imagine a Grade 4 cousin asks: “Why does the $\times 11$ trick work? Isn’t it just magic?”

Write 5–7 sentences explaining why it’s NOT magic. Use the words **distribute**, **digit sum**, and **place value**.

- 10 min individual writing.
- 5 min pair-sharing: read your paragraph to your partner; they tell you one part that’s clear and one part that’s unclear.

Wrap-up (5 min)

- Read aloud one strong paragraph (with permission).
- Preview Day 4: “Tomorrow — extend the trick to 3-digit numbers. The algebra will scale.”

Student-facing content

Why the $\times 11$ trick works

Write any 2-digit number as $10a + b$ — the first digit is worth 10, the second is worth 1.

Distribute the multiplication by $11 = 10 + 1$:

$$(10a + b) \times 11 = (10a + b) \times 10 + (10a + b) \times 1 = 100a + 10b + 10a + b = \mathbf{100a + 10(a + b) + b}.$$

In plain language: **hundreds = a , tens = $(a + b)$, units = b** . That’s the trick.

When $a + b \geq 10$, the “ $10(a+b)$ ” term overflows the tens place by exactly one hundred — that’s the carry.

It is not magic. It is the distributive law.

Homework

- Apply the proof to a number you choose. Pick any 2-digit number, write it as $10a + b$, distribute by 11, and show the final form. Show all 5 lines.

Differentiation

- **Tier up:** Prove the same identity for $\times 111$ (multiplication by 111). What’s the equivalent “split-and-add” pattern? Test it on 23×111 .
- **Tier down:** Don’t worry about the carry-case derivation. Get the no-carry version of the proof working.

Teacher notes

- This is the conceptual core of the module. Don’t rush.
- Students who can write the algebra but can’t *speak* it should be pushed to verbalise — “distribute” is the verb that matters here.
- The most common error in student paragraphs: confusing “the trick” with “the answer.” The trick is the algorithm; the algebra justifies the algorithm.
- About 20% of students will object: “*But this is just multiplication, not anything special.*” That’s the right reaction. Honour it: “*You’re right — and that’s the deeper point. The ‘special’ name (Ekādhikena Pūrveṇa) is a label; the math underneath is regular algebra.*”

Citation(s) used in this lesson

- (None required for the algebra itself — it is original derivation.)
- Optional teacher reference: Tirthaji (1965), sūtra 1, for the framing.

Day 4 — Three-Digit $\times 11$

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will compute any 3-digit $\times 11$ product using the running “neighbour-sum” pattern, including carry cases, and will complete Part A of the formative quiz.

Materials

- Whiteboard.

- Printed worksheet of 10 three-digit $\times$ 11 problems.
- Formative quiz Part A (quizzes/formative.md).

Lesson flow

Warm-up (5 min)

- Neighbour-sum chorus: 10 quick 2-digit $\times$ 11 problems including carry cases. (Should be near-instant by now.)

Core (18 min) — The 3-digit pattern

1. **Set up.** “Yesterday we proved $(10a+b)\times 11$. What happens with a 3-digit number $100a + 10b + c$?”
2. **Derive together** (or have students derive):

$$\begin{aligned}
 (100a + 10b + c) \times 11 &= (100a + 10b + c)(10 + 1) \\
 &= 1000a + 100b + 10c + 100a + 10b + c \\
 &= 1000a + 100(a + b) + 10(b + c) + c
 \end{aligned}$$

3. **Read the result.** “Thousands = a . Hundreds = $a + b$. Tens = $b + c$. Units = c .”

In words: **first digit stays, then each pair of adjacent digits sums and slides into the gap, then the last digit stays.**

4. **Worked example: 234×11 .**

Splits and gaps: 2 _ _ 4

↑ ↑

(a+b) (b+c)

$$\begin{aligned}
 2 + 3 &= 5 \\
 3 + 4 &= 7
 \end{aligned}$$

Answer: 2 5 7 4 $\rightarrow$ 2574.

Check: $234 \times 11 = 234 \times 10 + 234 = 2340 + 234 = 2574$. ✓

5. **Worked example with carry: 567×11 .**

$$\begin{array}{rcl}
 5 & _ & _ & 7 \\
 5 + 6 & = & 11 & \leftarrow \text{carry case} \\
 6 + 7 & = & 13 & \leftarrow \text{carry case}
 \end{array}$$

Strategy: work **right to left** to handle carries properly.

Units: 7.

Tens: $b + c = 6 + 7 = 13$. Write 3, carry 1.

Hundreds: $a + b + \text{carry} = 5 + 6 + 1 = 12$. Write 2, carry 1.

Thousands: $a + \text{carry} = 5 + 1 = 6$.

Answer: 6 2 3 7 $\rightarrow$ 6237.

Check: $567 \times 11 = 5670 + 567 = 6237$. ✓

6. **Generalisation.** For an n -digit number, the trick is the same pattern:
- Outer digits stay.
 - Every interior digit becomes the sum of the two original digits flanking that position.
 - Carries cascade from right to left.

Activity (15 min) — Worksheet

- 10 problems mixing 3-digit no-carry and carry cases:

$$\begin{array}{ccccc} 123 \times 11 & 234 \times 11 & 345 \times 11 & 456 \times 11 & 142 \times 11 \\ 365 \times 11 & 478 \times 11 & 529 \times 11 & 687 \times 11 & 818 \times 11 \end{array}$$

- Pair work, self-check by long multiplication on any 3.

Formative Quiz Part A (5 min)

- Distribute `quizzes/formative.md` Part A.
- 5 quick problems, mixed 2- and 3-digit $\times 11$. Closed-notebook.

Wrap-up (2 min)

- Preview Day 5: “*Tomorrow — speed day. We’ll retest your Day 1 problem (73×11) using the trick.*”

Student-facing content

3-digit $\times 11$

For a number $abc \times 11$: 1. Write $a_ _ c$ (outer digits unchanged). 2. Fill the first gap with $a + b$. 3. Fill the second gap with $b + c$. 4. Handle carries right to left.

Example: $234 \times 11 \rightarrow 2_ _ 4 \rightarrow 2 (2+3) (3+4) 4 \rightarrow 2574$.

The algebraic identity: $(100a + 10b + c) \times 11 = 1000a + 100(a+b) + 10(b+c) + c$.

Homework

- Solve 10 three-digit $\times 11$ problems on the back of today’s worksheet. Time yourself — goal: under 90 seconds total.

Differentiation

- **Tier up:** Try a 4-digit $\times 11$ problem. Derive the pattern first (the identity will give you 5 terms). Test on 2345×11 .
- **Tier down:** Stay on 2-digit $\times 11$ problems with carries; build speed there before adding the 3-digit layer.

Teacher notes

- The right-to-left direction for carry handling is the key procedural insight. Some students will try left-to-right and run into trouble; redirect them.

- For 567×11 , the cascading carries (13 in tens, 12 in hundreds) are dramatic. Use that example.
- The pattern is sometimes taught with “underline carries” notation; we use right-to-left because it generalises better to n -digit numbers.

Citation(s) used in this lesson

- *Vedic_Maths.pdf* (ISKCON Desire Tree) — practice problems.
- Tirthaji (1965), sūtra 1.

Day 5 — Mental-Math Speed Day

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will retest their Day 1 baseline problem (73×11) using the trick, complete a paired speed race, and finish Part B of the formative quiz.

Materials

- Day 1 baseline sheet (returned to each student).
- Stopwatch.
- `activities/activity-01-speed-race.md` worksheet.
- Formative quiz Part B (`quizzes/formative.md`).

Lesson flow

Warm-up (3 min)

- Quick chorus: 12 problems, mixed 2- and 3-digit $\times 11$. Cover the carry case.

Day 1 retest (5 min)

- Return Day 1 baseline sheet to each student.
- Re-solve 73×11 using the trick. Time yourself.
- Record:

Day 1 time (long multiplication):	_____	sec
Day 5 time ($\times 11$ trick):	_____	sec
Improvement:	_____	%

- Most students will see >70% reduction (e.g. 25 sec $\rightarrow$ 5 sec).

Activity (20 min) — Paired speed race

See `activities/activity-01-speed-race.md` for the full set-up.

- Round 1 (5 min): Solo, 20 mixed-difficulty problems. Each problem on a flashcard.

- Round 2 (5 min): Paired. Partner reads problem; you answer aloud.
- Round 3 (5 min): Group chorus — teacher displays 10 problems; class shouts the answer.
- Round 4 (5 min): Quick-fire — teacher calls a 2-digit number; first student to call back the $\times 11$ product gets a tally mark. (Light competition; no cumulative score across days.)

Formative Quiz Part B (15 min)

- Distribute Part B.
- 8 questions, mixed 2- and 3-digit $\times 11$, plus 2 short-answer “explain why” items.

Wrap-up (2 min)

- Preview Day 6: *“Tomorrow — the trick extends to $\times 12$, $\times 13$, ... all the way to $\times 19$. The middle-add step scales by a multiplier.”*

Student-facing content

Speed Day

Today’s question: how much faster has your mental $\times 11$ gotten in 4 days?

Record your Day 5 retest time. Compare with Day 1. The gap is the proof of practice.

By end of class you should hit roughly: - 2-digit $\times 11$ (no-carry): under 2 seconds - 2-digit $\times 11$ (carry): under 4 seconds - 3-digit $\times 11$: under 8 seconds

Homework

- 25-problem mixed drill sheet. Solo, untimed. Target: 100% correct.

Differentiation

- **Tier up:** Try 4-digit $\times 11$ problems. Sample: 1234×11 , 2345×11 , 4567×11 . Use the right-to-left carry-cascade method.
- **Tier down:** Focus only on the 2-digit drill today. Build fluency before adding 3-digit.

Teacher notes

- The Day 1 $\rightarrow$ Day 5 contrast is the emotional payoff of the module. Make sure every student records it; the personal evidence drives motivation.
- Quick-fire (Round 4) should NOT escalate into pure competition. Use it lightly; rotate “first responder” so no single student dominates.
- Some students will plateau on the carry case. Pair them with a peer-coach in Round 2 specifically.

Citation(s) used in this lesson

- (No new sources today — applied practice of Days 1–4.)

Day 6 — *Anurūpyeṇa*: Extending to $\times 12$, $\times 13$, ..., $\times 19$

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will apply Tirthaji's *Anurūpyeṇa* sub-sūtra ("proportionally") to multiply any 2-digit number by 12, 13, ..., 19, with at least 80% accuracy on a mixed worksheet.

Materials

- Whiteboard.
- `activities/activity-02-anurupyena.md` worksheet.
- Stopwatch.

Vocabulary introduced today

- *Anurūpyeṇa* (IAST: *anurūpyeṇa*; lit. "proportionally") — a sub-sūtra in Tirthaji's corpus. We use it as the name for the proportional scaling that extends the $\times 11$ pattern to $\times 12$, $\times 13$, ..., $\times 19$.

Lesson flow

Warm-up (5 min)

- Recall: state the $\times 11$ trick in one sentence.
- "Outer digits unchanged; middle is the digit sum."

Core (18 min) — The scaling pattern

1. **Question.** "What does $\times 12$ look like? Try 23×12 by long multiplication first."

$$23 \times 12 = 23 \times 10 + 23 \times 2 = 230 + 46 = 276.$$

2. **Derive.** "Now distribute $(10a + b) \times 12 = (10a + b)(10 + 2)$:

$$\begin{aligned} &= 100a + 10b + 20a + 2b \\ &= 100a + 10(b + 2a) + 2b \end{aligned}$$

Hmm — that's awkward. $2b$ in the units, $(b + 2a)$ in the tens. Let's organise it more cleanly.

3. **The Tirthaji pattern.** For N in $\{11, 12, \dots, 19\}$, write $N = 10 + k$ where $k \in \{1, 2, \dots, 9\}$. Then:

$$\begin{aligned} (10a + b) \times (10 + k) &= 100a + 10b \cdot k + 10ka + bk \\ &\quad \text{Wait — let's redo carefully:} \\ &= (10a + b) \times 10 + (10a + b) \times k \\ &= 100a + 10b + 10ak + bk \\ &= 100a + 10(b + ak) + bk \end{aligned}$$

Hmm — bk could be 2 digits. The pattern is: **scale each digit by k as you slide along.**

Easier form for students: For $\times N = \times(10 + k)$:

- Start with the leftmost digit.
- Multiply it by k , add the next digit, write the units of that sum (carry as needed).
- Continue right.
- The rightmost step: multiply the last digit by k .

This is the **Tirthaji form of the Anurūpyeṇa extension**.

4. Worked example: 23×13 . ($k = 3$.)

Set up: 0 2 3 (pad with 0 on left to track the carry-out)

Right to left:

- Units: last digit $\times k = 3 \times 3 = 9$. Write 9.
- Tens: middle digit $\times k + \text{next-right digit} = 2 \times 3 + 3 = 9$. Write 9.

- Hundreds: leftmost digit $\times k + \text{next-right} = 0 \times 3 + 2 = 2$.
Wait – we want the LEFTMOST as just itself with the carry...

(Pause — this is getting confusing. Let's use a different presentation.)

5. Cleaner presentation for $\times 12$ to $\times 19$.

For $N \times 12$ of a 2-digit number ab : simply use long multiplication mentally, but in the order:

- units of answer = $b \times 2$ (carry if ≥ 10)
- tens of answer = $a \times 2 + b$ (plus carry; carry if ≥ 10)
- hundreds of answer = a (plus carry)

For $N \times 13$: replace $\times 2$ with $\times 3$ everywhere.

General: For $\times(10 + k)$, multiply each digit by k and add the digit to its right (the “running scan”).

6. Examples (boxed):

23×12 :

units: $3 \cdot 2 = 6$
tens: $2 \cdot 2 + 3 = 7$
hundreds: 2 (no carry)
→ 276 ✓

23×13 :

units: $3 \cdot 3 = 9$
tens: $2 \cdot 3 + 3 = 9$
hundreds: 2
→ 299 ✓

45×13 :

units: $5 \cdot 3 = 15 \rightarrow$ write 5, carry 1
tens: $4 \cdot 3 + 5 + 1 = 18 \rightarrow$ write 8, carry 1

hundreds: $4 + 1 = 5$
→ 585 ✓

Check 45×13 : $45 \times 10 + 45 \times 3 = 450 + 135 = 585$. ✓

7. **The name.** “Tirthaji (1965) calls this proportional scaling **Anurūpyeṇa** — ‘proportionally.’ The $\times 11$ case is the special case $k = 1$, where the scaling is trivial.”

Activity (15 min) — Mixed drill

See activities/activity-02-anurupyena.md.

- 16 problems across $\times 12$, $\times 13$, $\times 14$, $\times 17$, $\times 19$.
- Paired self-check.

Wrap-up (2 min)

- “How does $\times 11$ fit into the $\times 12$ – $\times 19$ family?” (It’s the $k=1$ special case.)
- Preview Day 7: “What about $\times 23$? $\times 47$? When does the trick stop helping?”

Student-facing content

The $\times 12$ to $\times 19$ family — **Anurūpyeṇa**

For any $N = 10 + k$ (where k is 1 to 9), multiply a 2-digit number $ab \times N$ by: 1. Units = $b \times k$ (carry if ≥ 10). 2. Tens = $a \times k + b$ (plus any carry; carry if ≥ 10). 3. Hundreds = a (plus any carry).

When $k = 1$ (i.e. $\times 11$), this is the original “split and add” trick.

Tirthaji (1965) names this scaling pattern **Anurūpyeṇa** — “proportionally.”

Homework

- 12 problems mixing $\times 12$, $\times 13$, $\times 14$: see worksheet attached. Self-check with long multiplication on any 4.

Differentiation

- **Tier up:** Try 3-digit $\times 13$ problems. Use the right-to-left running-scan from Day 4 with the k -scaling from today.
- **Tier down:** Focus only on $\times 12$ today. Build fluency before adding higher k .

Teacher notes

- **Anurūpyeṇa** is the pattern-naming step. Some students will object: “This is just long multiplication done right-to-left.” They are correct. Honour the observation; reframe: “The win is doing it **MENTALLY** in one pass, with a clear rhythm.”
- The procedural complexity grows quickly with k . $\times 17$ and $\times 19$ are harder than $\times 11$ by a real margin. Calibrate the problem set accordingly.
- This day is genuinely harder than Days 1–5. If the class is shaky, slow down — skip the harder k values and revisit on Day 9.

Citation(s) used in this lesson

- Tirthaji (1965), sub-sūtra *Anurūpyeṇa*.
- *Microsoft Word - sutras1.pdf* — for the sub-sūtra list.

Day 7 — When the Trick Doesn't Help

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will identify which multiplications are well-suited to the *Ekādhikena/Anurūpyeṇa* family and which are not, and will articulate the conditions that make a different sūtra (*Nikhilam* or *Ūrdhva-tiryagbhyāṁ*) more appropriate.

Materials

- Whiteboard.
- 8-problem comparison worksheet ($\times 11$ trick vs. long multiplication vs. *Nikhilam*-style).

Lesson flow

Warm-up (5 min)

- Quick chorus: 10 problems from Day 6's $\times 12$ – $\times 19$ family.

Core (20 min) — Where the trick breaks down

1. **Try $\times 23$.** “Let’s try the same proportional pattern with multiplier $23 = 20 + 3$.”
 - For 14×23 , would the trick scale?
 - $(14)(20 + 3) = 280 + 42 = 322$. Long way works.
 - The “proportional middle” pattern doesn’t extend cleanly past 19 because the multiplier has two non-trivial digits.
2. **Why?** Recall the identity: $\times(10 + k) \rightarrow 100a + 10(ak + b) + bk$. Clean because the “tens” part of the multiplier is **just 1**. For $\times 23$, the algebra is:

$$\begin{aligned} & (10a + b)(20 + 3) \\ &= 200a + 20b + 30a + 3b \\ &= 200a + 30a + 20b + 3b \\ &= (100 \cdot 2 + 100 \cdot 0 + \dots) - \text{no longer a tidy "outer-stays, inner-sums" pattern.} \end{aligned}$$

3. **The honest summary.**

The $\times 11$ trick (*Ekādhikena Pūrveṇa*) and its scaled cousins (*Anurūpyeṇa* for $\times 12$ to $\times 19$) work because the multiplier is *very close to 10*. Specifically, the multiplier must be $10 + k$ where k is a single digit.

For multipliers farther from 10 ($\times 23$, $\times 47$, $\times 98$), a DIFFERENT sūtra is the right tool — Tirthaji’s *Nikhilam* (“all from 9, last from 10”) for multipliers near 100, or *Ūrdhva-tiryagbhyām* (“vertically and crosswise”) for general multiplication.

4. **Connect to Module 9 / 11.** Future modules will cover *Nikhilam* and *Ūrdhva-tiryagbhyām* in depth. Today we just note their existence.
5. **The bigger lesson.** “Every mental-math shortcut has a domain. Knowing *WHEN* a trick applies is part of being a fluent mathematician. The standard long-multiplication algorithm always works; the shortcuts are faster only inside their domain.”

Activity (15 min) — Comparison drill

- 8 problems. For each, decide *before* solving: would you use (A) the $\times 11$ family trick, (B) long multiplication, or (C) a different mental shortcut you can articulate?

1. 67×11	5. 84×23
2. 47×13	6. 99×99
3. 56×19	7. 25×12
4. 38×47	8. 102×11
- Pair work. Solve and discuss your method choice.
- Answers and “best method” given on the back.

Wrap-up (5 min)

- Class compiles “the $\times 11$ family domain”: *all 2-digit numbers $\times (11, 12, 13, \dots, 19)$* — about $90 \times 9 = 810$ distinct problems. That’s a significant mental-math territory.
- Preview Day 8: “*Tomorrow — where do these 16 sūtras come from? The honest history.*”

Student-facing content

The domain of the $\times 11$ trick

The “split and add” trick works for multipliers of the form $10 + k$ where k is a single digit (so $\times 11$ through $\times 19$).

For other multipliers, use: - Long multiplication (always works). - *Nikhilam* “all from 9, last from 10” — when both numbers are near 100 (e.g. 98×97). - *Ūrdhva-tiryagbhyām* “vertically and crosswise” — general multiplication.

Knowing **when to use** a trick is as important as knowing **how**.

Homework

- Find any 3 multiplication problems in your homework or a textbook. For each, decide which method ($\times 11$ family / long / other) you’d use and **why**. Write 1 sentence of justification per problem.

Differentiation

- **Tier up:** Find any 2-digit $\times$ 2-digit multiplication where the $\times 11$ trick is NOT optimal but a different mental-math shortcut would beat long multiplication. (Examples: $25 \times 24 = 25 \times 24 = 600$ via $25 \times 4 \times 6$; $75 \times 12 = 900$ via $75 \times 12 = 75 \times 12$.)
- **Tier down:** Focus on identifying $\times 11$ vs. non- $\times 11$ cases. Skip the “best alternative method” question.

Teacher notes

- This day is short by design — it’s the honesty hour. Don’t over-fill.
- Some students will resist the framing “*the trick has a limited domain*” and try to force-fit $\times 23$ into a pattern. Honour the impulse, but redirect: “*You’d be inventing your own technique — which is great, but it’s not what this trick does.*”
- Mention that Tirthaji’s 16 sūtras collectively cover a wide territory; $\times 11$ is just one of them. This sets up Day 8.

Citation(s) used in this lesson

- Tirthaji (1965), sūtra list (especially *Nikhilam* — sūtra 2, and *Ūrdhva-tiryagbhyām* — sūtra 3).
- *Microsoft Word - sutras1.pdf* — full sūtra list.

Day 8 — Tirthaji’s Sūtra System in Context

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will name at least 5 of Tirthaji’s 16 sūtras, explain in 2–3 sentences when and by whom *Vedic Mathematics* was published, and state honestly that the “Vedic” attribution is academically contested.

Materials

- Printed handout: the list of 16 sūtras + 13 sub-sūtras (from *Microsoft Word - sutras1.pdf*).
- Optional: a paragraph excerpt from Tirthaji’s preface (teacher’s prepared synopsis).

Lesson flow

Warm-up (3 min)

- Recall: name the sūtra behind the $\times 11$ trick. (*Ekādhikena Pūrveṇa*.)
- Name the sub-sūtra behind $\times 12$ – $\times 19$. (*Anurūpyeṇa*.)

Core (25 min) — The 16 sūtras and the provenance question

1. **The 16 sūtras.** Hand out the list. Walk through abbreviated:

#	Sanskrit	English gloss
1	<i>Ekādhikena Pūrveṇa</i>	“By one more than the previous”
2	<i>Nikhilam Navataścaramaṁ Daśataḥ</i>	“All from 9 and the last from 10”
3	<i>Ūrdhva-tiryagbhyāṁ</i>	“Vertically and crosswise”
4	<i>Parāvartya Yojayet</i>	“Transpose and apply”
5	<i>Śūnyaṁ Sāmyasamuccaye</i>	“When the sum is the same, that sum is zero”
6	<i>(Ānurūpye) Śūnyam Anyat</i>	“If one is in ratio, the other is zero”
7	<i>Sankalana Vyavakalanābhyām</i>	“By addition and by subtraction”
8	<i>Pūraṇāpūraṇābhyām</i>	“By completing or non-completing”
9	<i>Calana-Kalanābhyām</i>	“By differentiation and integration”
10	<i>Yāvadūnam</i>	“Whatever the extent of its deficiency”
11	<i>Vyaṣṭi-Samaṣṭiḥ</i>	“Specific and general”
12	<i>Śeṣānyaṅkena Caramena</i>	“The remainders by the last digit”
13	<i>Sopāntya-dvayam Antyam</i>	“The penultimate and the last”
14	<i>Ekanyūnena Pūrveṇa</i>	“By one less than the previous”
15	<i>Guṇitasamuccayaḥ</i>	“The product of the sum is the sum of the products”
16	<i>Guṇaka Samuccayaḥ</i>	“All the multipliers”

Plus 13 sub-sūtras (including *Anurūpyeṇa*).

2. Who was Tirthaji?

- **Bharati Krishna Tirthaji** (1884–1960). Śaṅkarācārya of Govardhana Maṭha (Puri) — i.e. head of one of the four major Hindu monastic seats founded by Ādi Śaṅkara.
- Studied at Madras Christian College. Multi-lingual scholar.
- Wrote the manuscript of *Vedic Mathematics* sometime between roughly 1911 and 1918 according to its editor’s foreword.
- The manuscript was lost or unfinished for decades.
- Tirthaji **died in 1960**.
- The book was **published in 1965**, five years after his death, edited by Vasudev Sharan Agrawala.

3. The provenance question. “Tirthaji claimed the 16 sūtras come from a Vedic text called the Atharva-veda Paṛiśiṣṭa — a supplement to the Atharva Veda.”

- **The problem:** No other Sanskritist has produced a manuscript of this *Paṛiśiṣṭa* containing these 16 sūtras.
- **The mathematician S. G. Dani** (TIFR, IIT Bombay) wrote a careful critique in 1993, reprinted in *Resonance* (October 2001), pointing out:
 - No manuscript evidence.

- The Sanskrit style of the sūtras is more recent than Vedic.
- The mathematical techniques have 19th–20th century parallels.

4. **Honest framing.** Two things can be true at once:

1. The $\times 11$ trick (and the rest of Tirthaji's 16-sūtra system) **works**. It is a useful, beautiful piece of mental arithmetic.
2. The trick is **not verifiably ancient Vedic mathematics**. The best current evidence is that Tirthaji synthesised it in the early 20th century, drawing on classical Indian arithmetic + his own pedagogical genius, and named the sūtras in Sanskrit.

The synthesis itself is a real Indian intellectual achievement. The “Vedic” naming is a 20th-century interpretive choice that scholars dispute.

5. **What's at stake.** If the sūtras are Vedic (3000+ years old), they're a major historical claim. If they're 20th-century synthesis, **they're still a real achievement** — just a different one. Tirthaji's pedagogical genius is not diminished by being recent.

Discussion (15 min) — Take a position

Pose two questions to the class. Take a brief show-of-hands; then take 4–5 student statements (no debate, just listening).

1. “Does it matter whether the trick is ‘ancient’ or ‘20th-century’? Why or why not?”
2. “If it turns out to be 20th-century, should we still teach it as ‘Vedic mathematics’? What should we call it?”

End by noting: “At Senior band you'll read Dani's critique in full and form a deeper position.”

Wrap-up (2 min)

- Preview Day 9: “Tomorrow — mixed-practice day plus project work. Friday is the final assessment.”

Student-facing content

Tirthaji and the 16 sūtras

Bharati Krishna Tirthaji (1884–1960), Śāṅkarācārya of Govardhana Maṭha, wrote *Vedic Mathematics* between roughly 1911 and 1918. The book was published in 1965, five years after his death.

Tirthaji claimed the 16 sūtras (including our *Ekādhikena Pūrveṇa*) come from a Vedic appendix-text. **No other Sanskritist has produced this manuscript.** S. G. Dani (1993) argues the sūtras are best understood as a 20th-century synthesis with Indian-mathematics roots, not a verifiably ancient Vedic teaching.

The trick works. The history is contested. Both can be true. The Indian intellectual achievement is real either way.

Homework

- Read the 16-sūtra list. Pick ONE sūtra (other than the two we've studied) and find what mathematical operation it's used for. (Sūtras 2, 3, 4, 10, 14 are well-documented in Tirthaji 1965.) Write 3 sentences.

Differentiation

- **Tier up:** Read a one-paragraph excerpt from Dani (1993). What is Dani's strongest critique? What does he NOT criticise?
- **Tier down:** Memorise the names of sūtras 1, 2, 3, and 14 (the four most commonly taught). Skip the historicity discussion details.

Teacher notes

- Some students (and their families) believe firmly that the trick is “ancient Vedic mathematics.” Honour the belief; don't dismiss it. Frame your position as “the academic consensus is more sceptical, and you'll read the evidence at Senior band.”
- Avoid civilisational chest-thumping (“ancient Indians knew this before everyone”). The Indian achievement here is the synthesis and clarity of Tirthaji's pedagogical work; that's enough to celebrate without overreach.
- Avoid the opposite error too — dismissing the trick as “just Western math with Sanskrit labels.” The pedagogical insight (the named patterns, the rhythm) is genuinely Tirthaji's.

Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. — Sūtra list and preface.
- Dani, S. G. (1993). “Myths and Reality: On Vedic Mathematics.” Reprinted in *Resonance* (October 2001).
- *Microsoft Word - sutras1.pdf* (SutraGanita corpus) — the 16+13 sūtra list.

Day 9 — Mixed Practice + Project Session

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will complete a mixed-practice drill covering 2-digit $\times 11$, 3-digit $\times 11$, and $\times 12$ – $\times 19$ problems, and will make substantial progress on their project (video, workbook, or proof poster).

Materials

- Mixed-practice drill sheet (20 problems).
- Project materials per pair: paper, markers, phone (for video pairs), template sheets (for workbook pairs).
- `assessments/rubric.md` posted on the wall for reference.

Lesson flow

Warm-up (3 min)

- 8 quick mental $\times 11$ problems including carry. Should be near-instant.

Mixed-practice drill (12 min)

- Distribute 20-problem sheet:
 - 5 problems: 2-digit $\times 11$ (no-carry)
 - 5 problems: 2-digit $\times 11$ (carry)
 - 5 problems: 3-digit $\times 11$
 - 5 problems: 2-digit $\times 12$ through $\times 19$
- Solo, untimed, but aim to finish in 10 min.
- 2 min: pair-swap, peer-check three problems each.

Project session (28 min)

- Project format options (pick on Day 8 or earlier):
 - **Format 1: 60-second teaching video** for a Grade 4 audience.
 - **Format 2: Printed workbook** — 25 problems organised in difficulty levels, with worked examples and an answer key.
 - **Format 3: Proof poster** — one A3 sheet showing the algebraic derivation of $(10a + b) \times 11 = 100a + 10(a+b) + b$ with examples and a 3-line explanation.
- Each pair shows the teacher their plan in the first 5 min. Teacher signs off or redirects.
- Work time. Teacher circulates:
 - Check video pairs have a quiet recording space.
 - Check workbook pairs have grid paper.
 - Check poster pairs have A3 sheets and at least 2 markers.
- Final 5 min: each pair shares their working title and ONE thing they're stuck on. Class offers one suggestion.

Wrap-up (2 min)

- *"Tomorrow is the final assessment and project share. Bring your finished artefact. Polish, don't restart."*

Student-facing content

Today's two jobs

1. **Drill.** 20 mixed problems. Self-assess.
2. **Project work.** Make real progress today — by end of class you should be 70% done.

Project Day 10: 2-minute presentation + 1-min Q&A. Be ready.

Homework

- Finish your project at home. 30 minutes max. If it's not finishable in 30 minutes, **simplify it** — better a complete simple project than a half-done ambitious one.

Differentiation

- **Tier up:** Add a “challenge” section to your project: a 4-digit $\times 11$ worked example with cascading carries.
- **Tier down:** Use the workbook template provided. Don't custom-design from scratch today.

Teacher notes

- The project here is intentionally smaller-scope than Module 5's, because the technique is narrower (just $\times 11$ plus the $\times 12$ – $\times 19$ family) and the assessment is more on the algebra than on the artefact.
- Watch for pairs who haven't started by mid-class. Pull them aside; offer the simplest format (proof poster) as a fallback.
- The proof poster format is recommended for pairs that struggled on Day 3 — making the algebra public forces them to internalise it.

Citation(s) used in this lesson

- (No new citations — application day.)

Day 10 — Final Assessment + Project Showcase

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate mastery of the $\times 11$ family of techniques via a summative quiz and a 2-minute project presentation.

Materials

- Summative quiz (`quizzes/summative.md`).
- Stopwatch.
- Rubric (`assessments/rubric.md`) per pair.
- Reflection slip per student.

Lesson flow

Warm-up (2 min)

- “Quiz first (15 min), then presentations.”

Part 1: Summative Quiz (15 min)

- Distribute `quizzes/summative.md`. Closed notebook.
- Covers: 2-digit $\times 11$ (with and without carry), 3-digit $\times 11$, *Anurūpyeṇa* extension, the algebraic identity, the historicity question (1 short-answer).
- Collect at 15-min mark.

Part 2: Project Presentations (25 min)

- 2-min presentation + 1-min Q&A per pair (3 min total per pair).
- ~8 pairs fit in 25 min. Reserve next class for remaining pairs if needed.
- During each presentation:
 - Teacher marks rubric in real-time.
 - One assigned audience member asks one question.
 - Other audience members fill “I learned / I wonder” slips.

Wrap-up (3 min)

- Closing question (oral): “*What’s one mental-math habit you’ll keep using after this module?*”
- Reflection slip submitted (reproduced in `student-workbook.md`).

Student-facing content

Today: quiz + presentations

15 min for the summative quiz (closed book).

2-minute presentation per pair, then 1-min audience question.

Reflection (due at end of class): 1. What was the most surprising thing you learned about $\times 11$? 2. Did your view of “Vedic mathematics” change during this module? How? 3. If you had three more days, what would you do differently?

Homework

- None. Module is complete.

Differentiation

- **Tier up:** Students finishing the quiz early can attempt a bonus “challenge” sheet — 4-digit $\times 11$ plus a 2-digit $\times 19$ mix.
- **Tier down:** Open-notebook quiz for students with documented learning needs.

Teacher notes

- Mark the quiz immediately after class while the presentations are fresh.
- Don’t rush the reflection — the “did your view change?” question is the cultural-honesty payoff of the module.
- If presentations spill into Day 11, protect that time.

Citation(s) used in this lesson

- All sources verified across the module.

Writing Prompts

Writing Prompts — Module 7 Eleven Multiplication (Middle)

Six prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

- 1. (Day 1 or Day 5, end of class, 3 min)** What did it feel like the first time you watched the teacher do 23×11 in 2 seconds? Describe what your mind did — disbelief, suspicion, “show me again”?
- 2. (Day 5 or Day 7, 5 min)** You’ve now seen one Vedic-math trick in detail. Has your *feeling* about mental math changed since Day 1? Be honest — even if the answer is “no.”

Explanatory prompts

- 3. (Day 3 or Day 4, homework, ~120 words)** Explain the $\times 11$ trick to a Grade 4 student who has never seen it. Use one specific example. No more than 6 sentences. Include the carry case in 1 sentence.
- 4. (Day 6 or homework, ~100 words)** Explain the *Anurūpyeṇa* extension in your own words. Why does the same idea scale from $\times 11$ to $\times 12$, $\times 13$, ..., $\times 19$? Use one example.

Argumentative prompt

- 5. (Day 8 or homework, 1 paragraph)** Some people claim the $\times 11$ trick is “ancient Vedic mathematics.” Others say it’s a 20th-century pedagogical technique published in 1965. Pick a position and defend it in 6–8 sentences. Cite at least one source.

Creative prompt

- 6. (Day 4 or Day 9, optional, ~150 words)** Write a short script (50–150 words) in which a frustrated student and a confident $\times 11$ -trick user have a conversation. The frustrated student raises one good objection (“this is just multiplication, why a special name?”). The confident user gives a clear, honest answer. Let the conversation be real — no “and then they magically agreed” endings.
-

Prompt-to-band notes

- For **Primary** band, prompts 1 and 6 work as oral dialogues; prompt 3 works as a drawing.
- For **Senior** band, prompt 5 deserves a full essay with citations from Tirthaji (1965), Dani (1993), and Plofker (2009).

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the project rubric — does the writing acknowledge the other side, cite a source, reach a clear position?
- Prompt 6: no grading. Read the best ones aloud (with permission).

Homework

Homework — Module 7 Eleven Multiplication (Middle)

Day 1 → Day 2

Task: Find any five 2-digit numbers (with digit-sum < 10) around your house — phone numbers, page numbers, prices. Compute each $\times 11$ using the trick. Bring the worked list.

Time: 10 min.

Day 2 → Day 3

Task: Eight mental $\times 11$ problems including the carry case: 76, 84, 85, 88, 93, 96, 67, 78 — each $\times 11$. Write only the answers. Time yourself; target under 30 seconds total.

Time: 5 min.

Day 3 → Day 4

Task: Apply the proof to a number you choose. Pick any 2-digit number, write it as $10a + b$, distribute by 11, and show the final form. Show all 5 lines of working.

Time: 10 min.

Day 4 → Day 5

Task: Solve 10 three-digit $\times 11$ problems (on the back of today’s worksheet). Time yourself — goal: under 90 seconds total. Self-check three with long multiplication.

Time: 15 min.

Day 5 → Day 6

Task: 25-problem mixed drill sheet (handed out in class). Solo, untimed. Target: 100% correct.

Time: 15 min.

Day 6 → Day 7

Task: 12 problems mixing $\times 12$, $\times 13$, $\times 14$ (worksheet attached). Self-check 4 with long multiplication.

Time: 15 min.

Day 7 → Day 8

Task: Find any 3 multiplication problems in your homework or a textbook. For each, decide which method ($\times 11$ family / long / other) you'd use and **why**. Write 1 sentence of justification per problem.

Time: 10 min.

Day 8 → Day 9

Task: Read the 16-sūtra list. Pick ONE sūtra (other than the two we've studied) and find what mathematical operation it's used for. Write 3 sentences. Confirm your project plan in writing (4–6 sentences).

Time: 20 min.

Day 9 → Day 10

Task: Finish your project. Polish, don't restart. Time your presentation in your head — must be 2 min.

Time: capped at 30 min.

Day 10 → end

Task: None. Module is complete. Optional: read ahead to Module 9 / 11 (Nikhilam, Ūrdhva-tiryagbhyām) if curious.